Low-Frequency RoPE Subspace for Near–Far Sparse Attention

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***Abstract*—Page-to-token sparse attention must retain useful candidates while keeping selection cheaper than the attention work it saves. We study a near–far cascade with 32 shared low-frequency RoPE coordinates for page and token scoring, regional page quotas, and final attention on selected tokens using original-dimensional queries, keys and values. Budget analysis shows that reserving far pages need not preserve far-token capacity. Qwen3-8B diagnostics illustrate regional-coverage trade-offs: regional quotas improve mean far-region retention relative to a single pool but reduce it in 43 of 128 correlated observations. Historical quality comparisons do not isolate representation effects, and separate timing studies show slower sparse decoding than dense attention. Establishing a beneficial operating point therefore requires evaluating the compact 4/128-protection implementation with matched quality and preparation-inclusive cost.**

***Index Terms*—sparse attention, shared representation, regional retrieval, long-context inference**

I. Introduction

Long-context benchmarks such as LongBench and RULER evaluate models on tasks with long input histories [9], [10]. Long-context decoding repeatedly accesses previously generated keys and values. Sparse attention limits this work by selecting a smaller token set, but selection introduces its own scoring, ranking and memory accesses. A page-to-token cascade first decides which pages may contribute tokens to refinement. The aim is to retain useful candidates within a fixed token budget at a retrieval cost below the attention work saved.

Page summaries approximate relevance with different scoring targets. Mean pooling measures an average key response, whereas sign-aware extrema bound individual unquantized token scores in the same coordinates; this bound can be loose. Neither score directly ranks pages by their summed attention weight. Following ClusterKV's token/page perspective [7], Figure 1(a) compares candidate access against a whole-page mass-rank reference on one profiled query: mean and extrema scoring miss 13 and 17 of the 32 mass-rank reference pages, respectively. These missed pages contain 59.65% and 26.46% of the reference pages' summed attention weight. With enough positive-probability candidates to fill the quota, refinement cannot recover unprotected tokens from rejected pages. Summaries must therefore be assessed by candidate access as well as scoring cost.

A global TopK budget lets high-scoring candidates from one region occupy slots without guaranteeing minimum capacity for the other. In Figure 1(b), separate near and far reservations raise mean far retention from .6759 to .7207 while lowering it in 43 of 128 correlated observations. Quotas act on the tail rather than the mean: the single pool drives far retention to zero for some queries, while the quota arm never falls below .2667 (arm minima, unpaired). This controllable coverage trade-off calls for explicit page and token capacities: reserving far pages alone need not preserve far tokens.

These observations lead to a cascade with two explicit controls. A fixed set of complete low-frequency RoPE pairs supplies the scoring coordinates for both stages; separate near–far page quotas determine regional access to token refinement. The coordinate diagnostic in Figure 1(c) motivates the compact view, although its unpaired random baseline does not isolate frequency from pair structure. Original-dimensional Q/K/V are retained for final attention, so approximation enters through token selection. We also consider an optional prefill-informed far gate. The historical layer variation in panel (d) motivates this extension, but uses a different single-query statistic; the gate is disabled by default and must be assessed independently.

We formulate shared-coordinate regional selection and analyze how page and token budgets jointly constrain retrieval. The evaluation examines task quality, regional coverage and execution cost. Qwen3 diagnostics show changes in coverage, but historical scores do not isolate either mechanism and sparse decoding is slower than dense attention. The compact 4/128-protection implementation still lacks matched quality–cost evaluation.

II. Related Work

Quest, SparQ and HISA use query-aware approximate selection to reduce the active attention set [1], [2], [3]. We adopt Quest's sign-aware page extrema for far-page scoring. AsyncTLS already combines a page-to-token cascade with coordinate selection, quantization, per-query normalization followed by GQA probability aggregation, and final attention over original KV [4]. However, these selectors rank candidates in a single pool, without guaranteed minimum capacity for either region. We study these existing components with complete low-frequency RoPE pairs shared across both stages and independent near–far page quotas. Our L2 path scores the full history before masking.

Reducing attention traffic also requires considering what the selector itself reads. Token-Picker estimates probabilities from partial key bits and schedules further accesses on demand [12]; our coordinate-based scorer instead retains full-dimensional KV and derives auxiliary scoring views. KV compression and placement methods address complementary storage and movement constraints [5], [6], [7]. A smaller scoring view in our design reduces neither original-KV capacity nor, by itself, complete-request latency.

III. Near–Far Selection in a Shared Subspace

To control both selection cost and candidate access, the cascade uses two explicit mechanisms: a shared scoring view and regional page quotas. One 32-coordinate map supports both L1 page screening and L2 token refinement, separating low-dimensional selection from original-dimensional attention (Fig. 2). The cascade operates only during decoding; prefill remains dense. L1 assigns separate page quotas to the near and far regions, so pages in one region do not compete for the other region's allocation. In the ordinary L2 path, full-history scoring precedes the retained-page mask and regional token selection. The selected token indices are merged and deduplicated with protected positions for attention using the original Q, K and V; recomputing retained logits does not recover omitted tokens.

A. Shared low-frequency coordinates

Both regions and both selection stages use the same coordinate map within each layer and KV head, while each query head is scored separately. For Qwen3-8B, we select post-RoPE coordinates $J = \{48, \dots, 63, 112, \dots, 127\}$ from each 128-dimensional query and key. The indices retain 16 complete low-frequency split-half RoPE pairs (Supplementary Fig. S1). With coordinate-selection matrix

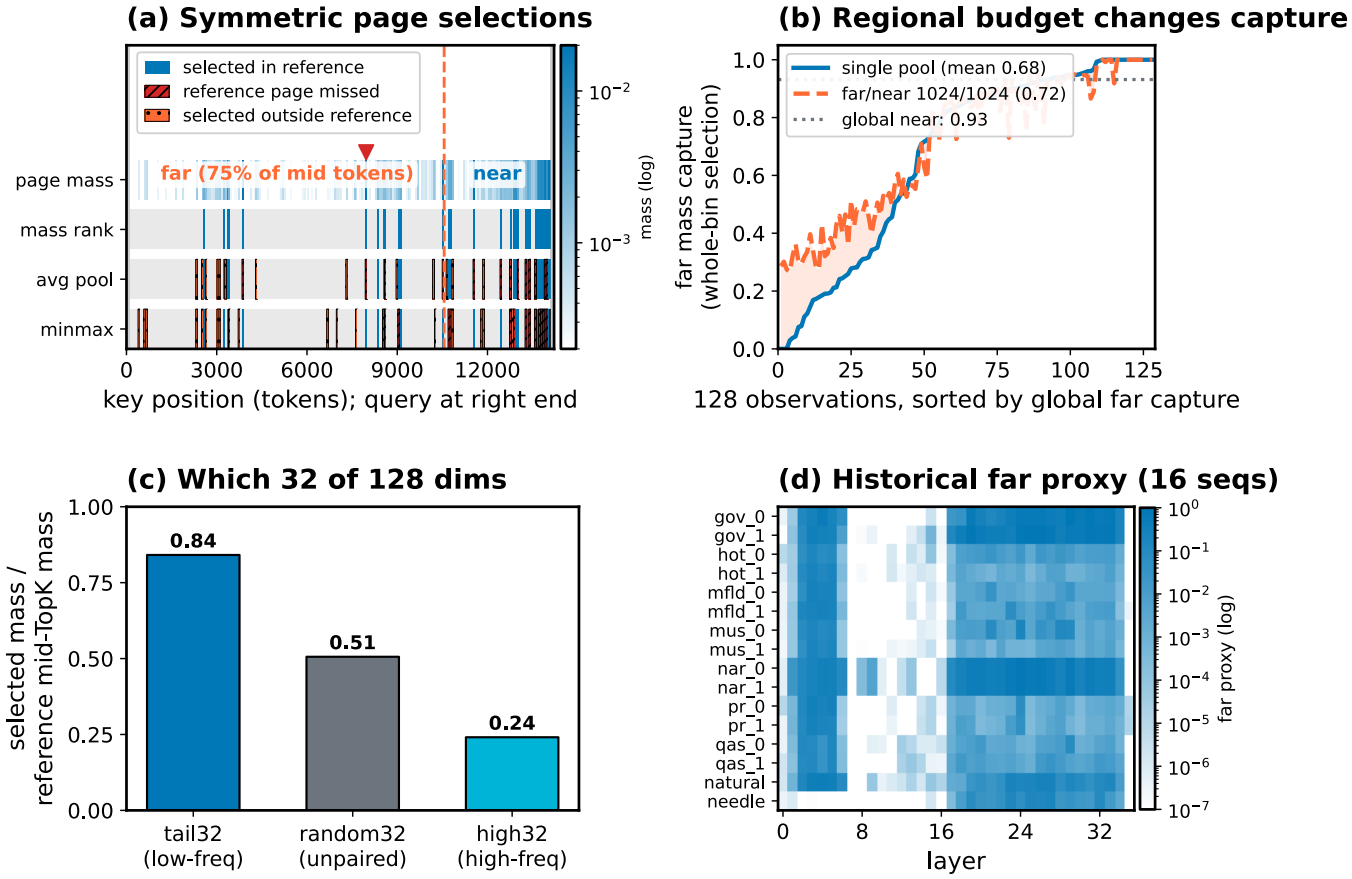


Fig. 1. Four motivating diagnostics; Qwen3-8B, 128/128 prefix/recent protection in (a)–(c). (a) MuSiQue sample 1, layer 24, query 14212. The top strip shows summed attention per page, not a selection. Below are the stored mass-rank, avg and minmax selections: 32 pages each, using 1989/1989/2048 tokens under a 2048-token cap. Blue denotes selected reference pages; red hatching denotes missed reference pages; orange dots denote selected pages outside the reference. Mass-rank is a whole-page comparison reference, not an optimal oracle, exact token TopK or DSA. Avg and minmax share 32 coordinates and head aggregation; the reference uses full-dimensional attention weights. Different regional allocations and realized token counts preclude inferring near/far specialization or task-quality ranking from these misses. (b) Four sequences contribute 128 correlated observations, ordered by global far retention. A global whole-bin cap of 2048 is compared with separate far/near caps of 1024/1024, without refill; red shading marks lower capture. Retention divides selected regional mass by full regional mass; the dotted line is global near retention. (c) Selected mass relative to full-dimensional reference mid-TopK mass; random32 does not preserve RoPE pairs. (d) The same 16-by-36 historical single-query proxy matrix gives 24/36 layer medians below .01 and 12–32 low-proxy layers per sequence. It uses the last sampled query, group-logit summation before softmax and far interval $[64, t-2048)$ (S6), unlike the current gate. These are diagnostic selections and profiles, not task-quality or decode-safety results.

$E_J \in \mathbb{R}^{128\times32}$, the scoring vectors are $z_q = E_J^\top q$ and $z_i = E_J^\top k_i$. Their slower positional rotation motivates this selection, although it does not guarantee preservation of important tokens. The compact implementation uses width-32 scoring tensors, whereas the historical quality runs used 32 active coordinates in width-128 L2 tensors (S7).

L1 aggregates the selected key coordinates into page summaries for coarse ranking, whereas L2 uses quantize–dequantize token keys for refinement. The L2 keys use 16 floating-point quantize–dequantize levels, not packed four-bit storage or INT4 arithmetic (S7). The two scoring views occupy separate buffers. Original keys and values remain full width. Preparing and maintaining the auxiliary views therefore adds work without compressing the original KV cache (S7).

B. Regional page and token budgets

In decoding, zero-based position t includes prefill, and $S = t+1$ counts valid cached positions through the current token, excluding padding. Let s and w denote the protected prefix and recent-window lengths. Their union and the unprotected middle are

$$\mathcal{P}_t = \{i : 0 \le i \le t,\ i < s \text{ or } i > t-w\}, \tag{1}$$

$$\mathcal{M}_t = \{0, \dots, t\} \setminus \mathcal{P}_t. \tag{2}$$

Overlapping protection counts once. Near is a suffix of $\mathcal{M}_t$; far is its complement there. Near excludes the protected recent window. Near uses fraction α of unprotected history, a one-page minimum and a downward page-aligned start; S7 specifies integer rules.

Regional page caps and token quotas constrain the cascade (Supplementary Fig. S2). Near pages use query–mean-key dot products to summarize page-level relevance; far pages use sign-aware extrema to bound unquantized shared-space token scores. This regional assignment is a heuristic. Summaries cover full physical pages, including protected boundary positions; L2 separately limits token regions (S7). For a nonempty page b, let $m_{b,j} = \min_{i\in b} z_{i,j}$ and $M_{b,j} = \max_{i\in b} z_{i,j}$. Quest's construction [1] gives

$$U_b(z_q) = \sum_j (z_{q,j}^+ M_{b,j} + z_{q,j}^- m_{b,j}), \tag{3}$$

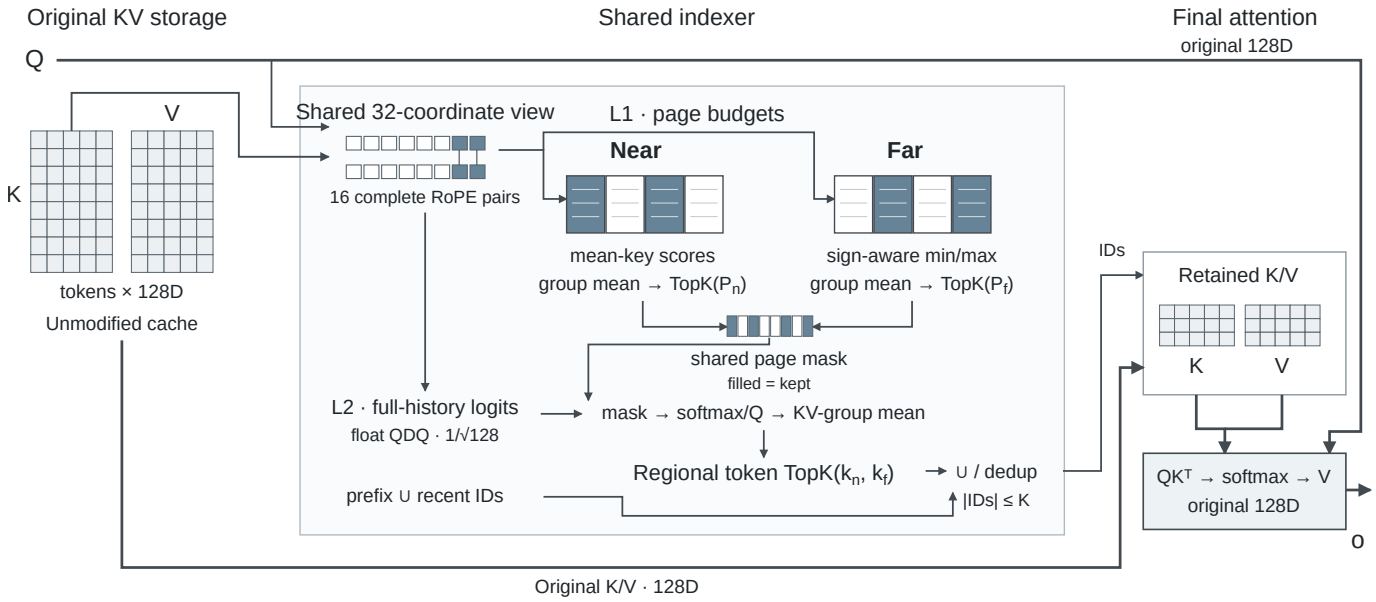


Fig. 2. Architecture of shared-subspace selection, with the optional gate inactive. A shared coordinate view feeds regional L1 page scoring and ordinary full-history L2 scoring. Page masks limit eligibility; per-query softmax and KV-group probability averaging precede regional token selection. Protected positions join deduplicated IDs within K. Near excludes the protected recent window. IDs define support over original-width K/V; physical gathering or dense masked computation depends on the backend. Original Q follows a separate path. Arrays are schematic.

where $z^+ = \max(z, 0)$ and $z^- = \min(z, 0)$. Each signed coordinate contribution is bounded by its chosen extremum, so $z_q^\top z_i \le U_b(z_q)$ for every unquantized shared-space key in b. This does not bound original-dimensional logits or independently quantized scores. Extrema may come from different tokens, making the bound loose enough to misrank pages and miss important tokens. Average-key scores lack this per-token bound.

For total cap $K \ge |\mathcal{P}_t|$, define the middle-token budget $K_{\text{mid}} = K - |\mathcal{P}_t|$. α controls near extent, β its share of L1 page cap K_1, and γ its nominal L2 quota from retained-page capacity. For the partitioned fixed-quota path (far active; no overrides), assume integer $K_1 \ge 1$, $\alpha > 0, 0 < \beta \le 1$ and $\gamma \ge 0$. With page size B and sufficient regional pages/tokens, $P_n = \max(1, \text{round}(\beta K_1))$, $P_f = \max(0, K_1 - P_n)$, $k_n = \min(\lfloor \gamma P_n B \rfloor, K_{\text{mid}})$ and $k_f = \max(0, K_{\text{mid}} - k_n)$; rounding uses ties-to-even. Thus $P_f > 0$ does not imply $k_f > 0$: page and token access are separate constraints. For $K_1 = 128$ and $\beta = .125$, L1 reserves 16 near and 112 far pages. With enough positive-probability candidates, L2 cannot recover unprotected tokens from L1-rejected pages. S7 details scarce-support and masked zero-tie exceptions.

With disjoint protection and sufficient candidates, the compact configuration ($K = 1024$, $s = 4$, $w = 128$) leaves 892 middle slots per KV head. Here $\gamma = .625$ yields $16 \cdot 64 \cdot .625 = 640$ near and 252 far slots (Supplementary Fig. S2). The partial prefix page can reselect up to four protected tokens, so 252 is nominal rather than guaranteed unprotected far coverage.

C. GQA selection and original-dimensional attention

Grouped-query attention (GQA) assigns four of Qwen3's 32 query heads to each of its eight KV heads. L1 averages their page scores before regional TopK. Ordinary L2 computes each query's full-history logits, applies the shared page mask, then softmax. Group-averaged probabilities determine regional token TopK. L2 uses scale $1/\sqrt{128}$ before softmax even with 32 scoring coordinates. Because softmax precedes group averaging, this rule generally differs from applying softmax to group-averaged logits. Width-32 scoring narrows each dot product; a smaller L1 shortlist restricts eligibility but does not reduce ordinary L2's full-history dot-product count.

For KV group h, let $\mathcal{S}_{t,h}$ be the deduplicated union of protected positions and regional winners, with $|\mathcal{S}_{t,h}| \le K$. Each query g uses this support for its own full-dimensional attention ($d = 128$):

$$o_{h,g} = \sum_{i \in \mathcal{S}_{t,h}} \frac{e^{q_{h,g}^\top k_{i,h}/\sqrt{d}}}{\sum_{j \in \mathcal{S}_{t,h}} e^{q_{h,g}^\top k_{j,h}/\sqrt{d}}} v_{i,h}. \tag{4}$$

Each query applies one softmax over the merged indices, avoiding the addition of separately normalized regional outputs [8]. Full-dimensional retained logits still use a restricted denominator, which can change the output. These indices define a shared logical support. Physical KV gathering and reuse depend on backend and length; short sequences can use a dense masked path.

D. Prefill-informed far gating

The optional gate tests whether low far-region attention during prefill can identify layers where far selection may be omitted during decoding; it is disabled by default. For each input and layer, a mean far-region attention proxy below ϵ sets the gate OFF; otherwise far selection stays active. Calibration uses a token-defined region that can differ from the page-aligned skipped region (S5). The first decode-path call freezes the decision until input reset (S5). OFF skips far L1/L2 scoring and selection in the supported fixed-quota path, refilling near L2 within unchanged near pages. Removing far support can change softmax denominators and group-averaged near rankings (S5). Representation preparation, sequence-wide score buffers and normalization remain.

The gate supports one sequence; the skip path requires valid length and nonempty far history (Supplementary Fig. S3; S5). The prefill

TABLE I
Compact design parameters and execution policy. These settings define the proposed implementation, not the historical runs. K includes unique protected tokens.

Parameter	Setting
Final token cap K	1024
Page size; L1 page cap K_1	64; 128
α, β, γ	.25, .125, .625
L1/L2 active coordinates	32/32
Prefix / recent protection	4/128
Prefill; decode layers	dense; all sparse
L2 scoring support	full history before mask

threshold does not bound decoding error; gate quality and total cost require separate evaluation.

IV. Experimental Setting

We evaluate three aspects separately: task quality, regional candidate coverage and execution cost. Table I defines the compact Qwen3-8B design with dense prefill, sparse decoding and a 1024-token cap including four prefix and 128 recent tokens. The available LongBench measurements instead use an earlier shared-subspace implementation: 128/128 protection and 32 active coordinates stored in zero-filled width-128 L2 keys. The comparison labels "full-dimensional page scoring" and "full-dimensional token scoring" denote reported variants whose complete execution configurations are unavailable. We retain these nominal labels without treating the runs as controlled single-stage ablations. Synthetic selector timing and model-level decode timing come from separate studies; none supplies matched quality and cost for the compact design. S6/S7 give configuration and measurement details.

Workloads and comparisons. The quality evidence comprises five LongBench tasks and a historical 11-task RULER [10] 128K comparison. The LongBench scores compare three nominal sparse scoring configurations; dense FullKV is a separate reference in the RULER and decode-timing records. Controlled sparse comparisons must align final unique-token cap K (including protection), model, inputs and phase policy. Realized token counts and head-sharing granularity also affect KV traffic; internal page budgets need not match (S7).

Metrics. Quality is the unweighted mean of task scores; task-specific changes are reported where available. Regional retention in Fig. 1(b) is selected attention mass divided by full regional mass; the coordinate diagnostic in Fig. 1(c) instead normalizes selected attention mass by full-dimensional reference mid-TopK mass. Selector time covers a synthetic single-layer operation, whereas synchronized TPOT measures model-level intervals after the first output token. Neither measures complete-request latency; evaluating that quantity requires accounting for preparation, maintenance and input/output work across the request. Because the quality and timing studies are unpaired, they cannot define a quality–latency frontier. Five task means also do not establish document-paired non-inferiority; that requires matched predictions and a prespecified acceptable-loss margin.

V. Evaluation and Discussion

A. Q1: Does reduced scoring width preserve task quality?

The five-task means sit within a descriptive spread of 0.33 points across the three nominal scoring configurations. On HotpotQA, Qasper, MuSiQue, GovReport and RepoBench-P [9], shared-subspace scoring scores 47.588, the nominal full-dimensional page variant 47.910 and the nominal full-dimensional token variant 47.666 (Supplementary Fig. S4). The shared-subspace score is lower by 0.322 and 0.078 points, respectively. These differences describe the recorded configurations; they do not show that reducing scoring width preserves quality. The variants are not verified single-factor interventions, and the shared-subspace run uses the earlier 128/128-protection implementation. Consequently, neither stage's representation effect nor the compact 4/128 design's quality can be inferred from this comparison. All three configurations are sparse; dense FullKV is a different reference.

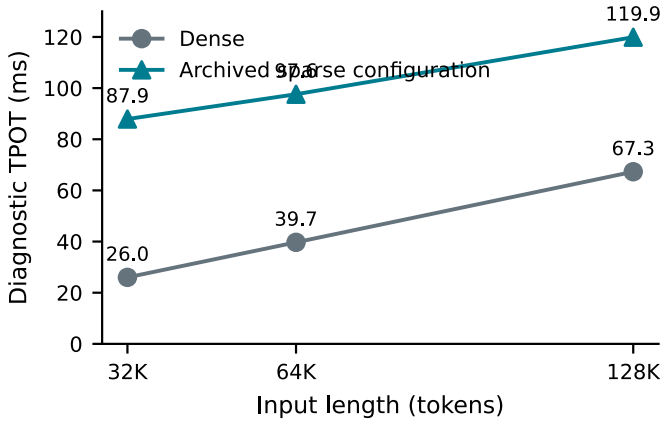


Fig. 3. Author-reported model-level decode TPOT for an archived sparse configuration and dense FullKV, Qwen3-8B with dynamic cache. Fixed 128-token output gives 127 after-first intervals, including device synchronization and argmax. The archived $(.125, .375, .625)$ setting differs from the selected 4/128 candidate. These medians do not measure complete-request E2E or establish a matched quality–cost comparison; provenance limits are in S6.

B. Q2: How do regional budgets constrain candidate access?

Regional reservation changes which distant tokens remain reachable. In Figure 1(b), separate quotas increase mean far retention but decrease it for 43/128 observations, illustrating the cost of redistributing a fixed budget. This diagnostic uses whole-bin selection on four sequences, so it does not measure production-selector task quality. The two-stage budget equations identify a second constraint: even a retained far page contributes no nominal far-token slots when the near quota exhausts K_{mid}. Substituting the historical RULER setting ($\beta = .375$, $K_1 = 128$, $B = 64$, $\gamma = .625$, protection 128/128) gives $K_{\text{mid}} = 768$ and $k_f = 0$. Figure 4 reports scores of 23.98 versus 46.43 for dense attention, a 22.45-point loss, but the algebra alone neither verifies its executed path nor attributes the loss to this quota. The current compact candidate lacks a matched dense comparison.

The protection-only configuration scores 26.09 against a 47.91 reference, while using 256 rather than 1024 tokens. The difference therefore combines retrieval-policy and budget effects.

C. Q3: Does compact scoring reduce execution cost?

Model-level decoding. The reported sparse configuration is slower than dense FullKV at every measured length (Fig. 3). Sparse TPOT medians are 87.9, 97.6 and 119.9 ms at 32K, 64K and 128K, compared with dense medians of 26.0, 39.7 and 67.3 ms. The corresponding ratios are 3.38, 2.46 and 1.78. Although the relative penalty decreases with length, absolute sparse TPOT increases. Thus these measurements show no decode-time benefit from that implementation. They include device synchronization and argmax, exclude tokenization, input transfer and text output, and use a configuration different from the compact 4/128 design. Aggregate medians cannot

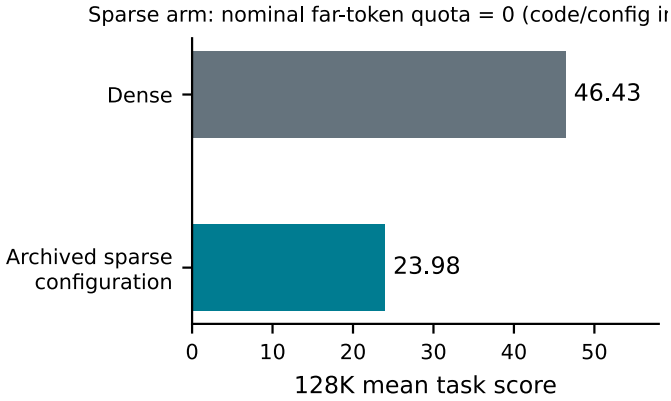


Fig. 4. Author-reported RULER 128K scores: 11 tasks, 100 examples per task and arm. The archived sparse configuration scores 22.45 points below dense FullKV; their macro-average task-score ratio is 51.6%. Reported $\beta = .375$, $K_1 = 128$ and 128/128 protection imply a nominal far-token quota of zero under the inspected formula (middle budget 768; S6). This is a code/configuration inference, not verified execution or causal attribution. The historical setting does not validate the selected 4/128 candidate or support sample-paired inference.

locate the cause of the slowdown or predict the compact implementation's performance; repetition and execution-path details are in S6.

Local selector cost. A separate two-round synthetic experiment (single layer, batch one, one query, 64K QKV) reports steady-state times of 1.3502 ms for nominal full-dimensional page scoring, 1.3598 ms for shared-subspace scoring and 1.4634 ms for nominal full-dimensional token scoring. The shared-subspace selector thus has no lower reported time than the nominal page-scoring variant in this setting. An unresolved sampling-protocol discrepancy limits interpretation of small differences (S6). These timings neither locate a bottleneck nor pair with LongBench quality or complete-request cost.

D. Optional gate: prefill is not a decode guarantee

Prefill statistics also fail to bound later far-region demand: an observed input–layer pair with prefill far-attention weight sum below .05 reaches .1678 during decoding. Relative to the reported reference score of 47.910, gate thresholds of .05 and .10 are associated with mean decreases of .034 and .218 points, respectively; three of five tasks decline in each configuration, including HotpotQA decreases of 1.05 and 1.26 points. The larger per-task losses demonstrate that a small average change can conceal task sensitivity.

The gate summaries omit the complete scoring configuration, precluding attribution to gated shared-subspace selection (S6).

Design implications and limitations. Three constraints emerge. First, scoring width controls the representation used for ranking, but the current L2 path still performs full-history scoring and normalization. Second, regional page access and regional token access require separate budgets; satisfying the first does not ensure the second. Third, the cost of preparing and using a selector must be recovered by reduced attention work. The reported timings do not show that recovery. The missing matched compact evaluation prevents a quality–cost conclusion, and the historical gate summaries do not identify the shared-subspace configuration. The evidence therefore supports the mechanism and these constraints, while leaving the proposed combination's comparative benefit unresolved.

VI. Conclusion

We studied sparse attention as a joint representation and allocation problem. A shared low-frequency coordinate map supplies page and token scoring, regional quotas control access to refinement, and final attention uses original-dimensional Q/K/V. The analysis distinguishes low-dimensional ranking from reduced execution work, and far-page reservation from final far-token coverage. Qwen3 diagnostics illustrate these trade-offs, including regional-retention losses, substantial historical quality loss and slower sparse decoding. Establishing a useful operating point for the compact 4/128 design requires measuring quality and preparation-inclusive cost under the same configuration; the available separate studies do not yet establish that benefit.

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